CHAPTER 15

DIMENSIONS OF PILLARS AND CHOICE OF MATERIALS

Pillars[1]

1 I now present, along the lines of the other treatises[2] but fully and in ordered fashion, the heights of the pillars, their diameter, form, decoration . . . etc.

2 *Sthāṇu*, *sthūṇa*, *pāda*, *jaṅghā*, *caraṇa*, *aṅghrika*, *stambha*, *talipa* and *kampa* are all synonymous (and designate pillars).[3]

Dimensions of the pillars

3-5a The ground floor pillars of a twelve storey building are eight and a half cubits high; by subtracting one span for each storey a height of three cubits is obtained for the pillars of the top storey. The height of a pillar may otherwise be calculated from a given height: it is double that of the corresponding base but the Svayambhū ordains that the height of a pillar be more than double that (of the base).[4]

1. Cf. *Mānasāra* (chapter 15), *Ajita* (chapter 14, Dagens 1984 pp.54-63), *Īśānaśiva* (Kriya. 31.19-74), *Śilparatna* (21.42 sq.) and G. Jouveau Dubreuil, *Dravidian Architecture*, 1917 [reprint 1972]. In this chapter *stambha* and its synonyms (cf. v. 2) designate pillars (or pilasters, see below); elsewhere in the text they are very often used as substitutes to *stambhavarga* (etc.) ('level of pillars', see chapter 14 note 9). Most of the prescriptions given here concern pillars as well as pilasters; in rare cases those last are specifically mentioned (see e.g. *kuḍyastambha*, v. 7).
2. *Lakṣaṇāntaratah*: "according to other (treatises dealing with the charasteristics (of pillars)".
3. For similar lists, see *Mānasāra* (15.4-6) and *Īśānaśiva* (Kriya. 31.19). *Kampa* is not, in the present case, the word which designates a kind of moulding (a 'fillet'), but a re-sanskritized form of Tamil term *kampam* (itself derived from skt. *skambha*!); it is not used in the sense of 'pillar' elsewhere than here and in 26.132 where is found the compound *kampadvāra* ('door attached to a pillar').
4. The *Svayambhū* is most probably the *Svayambhuvāgama*, one of the main canonical texts (*mūlāgama*) of the śaiva school; this is a very rare case of reference to a specific treatise. The proportions given here contradict those given in 14.16 and 47 (according to which the height of the base is to be less than half that of pillars.

[17]तदर्धं वा त्रिभागोनं[18] चतुर्भागोनमेव[19] वा ।
कुड्यस्तम्भस्य विस्तारस्तेन द्वित्रिचतुर्गुणः ॥ ७ ॥
पञ्चषड्गुण एवं स्याद्[20] भित्तिविष्कम्भ इष्यते ।
[21]जन्मोर्ध्वे स्तम्भविक्षेपहोमाः[22] स्तम्भविधौ विदुः ॥ ८ ॥

[स्तम्भभेदाः]

प्रतिस्तम्भः प्रतेरूर्ध्वे[23] चोत्तराधो हितायतिः ।
[24]जन्मोर्ध्वे स्तम्भनिक्षेपः स्तम्भायामस्त्रिभागभाक् ॥ ९ ॥
[25]गाम्भीर्यमवटं कृत्वा तदुपज्ञतलं[26] कुरु ।
[27]पादुकाद्युत्तरान्तस्थो निखातस्तम्भ इष्यते[28] ॥ १० ॥
[29]अधिष्ठानोत्तरान्तस्थो झषालस्तम्भ उच्यते[30] ।
तद्व्यासादर्कभागाद्यं[31] षड्भागोनाग्रविस्तरः ॥ ११ ॥
मूलभूस्तम्भतुङ्गस्य द्वादशाद्याः षडंशयुक्[32] ।
ऊर्ध्वोर्ध्वस्तम्भतुङ्गं स्यादेवं तद्विस्तरक्रमः ॥ १२ ॥

17. F : omits 7.
18. D : त्रिभागेन; E : त्रिभागं वा
19. D : भागेनमेव
20. D, E : वा
21. A : जन्मोर्धे; F : जनोर्धे
22. D et E : निक्षेपहोमाः
23. D : व्रतेरुर्ध्वे
24. A : जन्मोर्धे
25. C : गम्भीरमबटं कृत्वा तदुपर्यानतं कुरु ।
26. D : तदुपज्ञातरं
27. E : पादुकान्युत्तरान्तस्थो; F : पादुकादुत्तरान्तस्थो
28. F : उच्यते
29. C : omits 11a.
30. D : तुषालस्तम्भसिष्यते
31. A : भागाद्यु
32. D : षडंशकम्; F : पडंशकृत्

5b-8 The diameter of the ground floor pillars (of a twelve storey building) is twenty-eight digits; by subtracting two digits for each storey six digits are obtained (for the diameter of the pillars of) the top-storey. The diameter of a pillar may otherwise be one-tenth, one-ninth, or one-eighth of its height; that of the pilaster is half, two-thirds or three quarters (that of the pillar). It is said, as well, that the thickness of the wall is twice or three, four, five or six times (the diameter of the pilaster). It is said in the rule for pillars that the *homa* (?) which are above the plinth project all around the pillars (which rest on them)(?).[5]

Different types of pillars

9 The *pratistambha* rises from the stereobate'(which tops the base), up to the architrave. The part of the (*nikhāta*)-*stambha* above the plinth (of the base and below its summit) is equal to a third of the height of the pillar (?).[6]

10a After a deep excavation has been made the nature of the soil should be determined.[7]

10b-11a It is said that a 'pillar (whose bottom) is concealed' (*nikhāta-stambha*) rises from the plinth of the base up to the architrave. It is said that a *jhaṣālastambha* rises from the base up to the architrave.[8]

11b-12 The diameter of a pillar at its summit is from a sixth to a twelfth less than at its bottom. (It is also said) that the diameter of a ground floor pillar of a twelve storey building is a sixth of the height of that pillar and that the same relation exists between the height and the diameter of the pillars (of the) upper (storeys).

5. These pillars are probably *nikhātastambha* (below note 8). About *homa* see chapter 14 note 8.
6. If our interpretation is right, this implies that the base is higher than the level of pillars.
7. See chapters 4 and 14 (v. 2) but this half-verse seems to be out of place.
8. *Nikhātastambha*: the lower part of these pillars is sunk into the base of the building; it is the case of gateway pillars (see 24.29-31); door-jambs too seem to pertain to that category (cf. e.g. below chapter 30 v. 30b). According to Ganapati Sthapati *jhaṣālastambha* may be similar to those of the front veranda (poyal) of tamil houses which are often potbellied like fishes.

[33]अग्राकारं युगाश्रं तु कुम्भमण्डिसमन्वितम्[34] ।
ब्रह्मकान्तं तदष्टाश्रं विष्णुकान्तमिहोच्यते ॥ १३ ॥
[35]षडश्रमिन्द्रकान्तं स्यात् सौम्यं तत् षोडशाश्रकम् ।
कर्णमात्रेण तन्मूले चतुरश्रमितोर्ध्वतः[36] ॥ १४ ॥
अष्टाश्रं वा द्विरष्टाश्रवृत्तं पूर्वाश्रमीरितम्[37] ।
कुम्भमण्डियुतं[38] वाऽपि रुद्रकान्तं सुवृत्तकम् ॥ १५ ॥
विस्तारद्विगुणं मध्येऽष्टाश्रयुक्तं युगाश्रकम् ।
[39]वियुक्तं कुम्भमण्डिभ्यां मध्येऽष्टाश्रं तदुच्यते ॥ १६ ॥
चतुरष्टाश्रवृत्ताभं रुद्रच्छन्दं[40] समांशतः ।
दण्डाध्यर्धे द्विदण्डेनोत्तुङ्गद्विगुणविस्तृतम्[41] ॥ १७ ॥
पद्मासनं तु कर्तव्यं मूले पद्मासनं भवेत् ।
यथेष्टाकृतिसंयुक्तमूर्ध्वतो वा समण्डितम् ॥ १८ ॥
[42]चक्रवाकाकृतिव्याप्तं मूले पद्मासनान्वितम् ।
सभद्रं मध्यभागे तु भद्रकं तद् द्विमण्डितम्[43] ॥ १९ ॥
व्यालेभसिंहभूतादिमण्डितं यत्तु मूलतः ।
यथेष्टाकृतिसंयुक्तं तत्तन्नाम्ना समीरितम् ॥ २० ॥
[44]आत्तमेव तदायामे [45]शुण्डभेदसमन्वितम् ।
युतं तत् कुम्भमण्डिभ्यां[46] शुण्डपादमिति[47] स्मृतम् ॥ २१ ॥

33. E : omits 13-23.
34. D : संयुतम्
35. F : omits 14a.
36. C : इतोऽधरः
37. D : पूर्वाग्रमिष्यते
38. D : कुम्भमण्डयुतं
39. D : सुवृत्तं
40. A, C, F : रुद्रच्छन्द-; B : रुद्रच्छन्दः
41. D : विस्तृतम् for विस्तृतम्
42. D : omits 19-20.
43. C : द्विमण्डिकम्
44. D and F : वृत्तमेव
45. A : शुण्डु -; D : शुण्डि-
46. D : कुम्भमण्ड्याभ्यां
47. A : शुण्डुपादमिति; D : शुण्डिपादमिति

13-16 A pillar whose section is square (from) top (to bottom) and which is provided with a bell capital and an abacus is called *brahmakānta*;[9] if it is octagonal it is called *viṣṇu-kānta*;[10] if it is hexagonal it is called *indrakānta* and if it is sixteen sided, *saumya*. If a pillar is double section, square at the bottom and octagonal, sixteen sided or circular above, it is called *pūrvāśra* (but) it is called *rudrakānta* if it has a bell capital and an abacus and if it is completely circular.[11] If the median part is twice as high as it is wide and is octagonal section, if (the upper and lower parts) are square and if there is neither bell capital nor abacus it is referred to as a pillar 'octagonal in the middle'.

17a The *rudracchanda* pillar comprises three equal parts, one square, one octagonal and one circular.[12]

17b-19 The width of a lotiform base of pillar is one and a half or two modules or twice its height. The upper part (of a *bhadraka* pillar) is of whatsoever form desired or it comprises an abacus (and is) decorated with *cakravāka* (birds); it has a lotiform base at the bottom and projecting elements on its median part; this is a *bhadraka* (pillar) which has two abacus.[13]

20-22a A pillar which, no matter what the form of its upper part, is decorated from the bottom up with *vyāla* or elephants, lions, dwarves (etc. . . .),

9. *Kumbhamaṇḍisamanvita*: see v.30 and note *ad loc.* For the types of pillar described here see *Mānasāra* (15.20 sq.) and Dagens 1984 pp. 61.62 (descriptions of thirteen specific types according to *Ajita* 14.15b-32). A comparative chart of pillar types mentioned in agamic texts is given by N.R. Bhatt in his *Ajitāgama* edition (vol.1, between pp. 96 and 97).
10. According to *Ajita*, the drum of *viṣṇukānta* pillar may be octagonal or circular.
11. Its upper drum has a sixteen sided section according to *Ajita*; in the same text the *pūrvāgrastambha (Mayamata: pūrvāśra)* has a square base and a facetted circular drum.
12. That type of pillar comprises the Liṅga's three specific sections.
13. *Dvimaṇḍita*: the term *bhadra* is applied to any projecting element such as a forepart (e.g. 19.33) or a moulding. Here this projecting element is probably similar to an abacus (*maṇḍi*), hence the expression 'having two abacus'. It would have been possible to consider *dvimaṇḍita* as a mistake for *vimaṇḍita*; thus we would get here an alternative according to which the *bhadraka* pillar may be devoid of abacus; see however *Īśānaśiva* (Kriya. 31.24) where we find a similar expression in the description of a *bhadrakānta* pillar:

मूले पद्मासनश्चाथ चक्रवाकैरलङ्कृत ।
सभद्रमध्यभागश्च भद्रकान्तो द्विमण्डिकः ॥

[48]मुक्तोत्करणकर्माङ्गं पिण्डिपादं तदेव हि ।
कर्मायामेन चाग्रे तु चतुरश्रसमन्वितम्[49] ॥ २२ ॥
[50]तदधस्त्वर्धदण्डेन पद्मं वस्वश्रसंयुतम् ।
तदधस्तु विकाराश्रं[51] दण्डेनाब्जं तु पूर्ववत् ॥ २३ ॥
[52]तदधो दण्डमानेन मध्यपट्टं[53] युगाश्रकम् ।
पद्मं च षोडशाश्रं च पूर्ववत् परिकल्पयेत् ॥ २४ ॥
[54]मूले शेषं युगाश्रं स्याच्चित्रखण्डं तदुच्यते ।
तदेवाष्टाश्रकं मध्यपट्टं[55] श्रीखण्डमुच्यते[56] ॥ २५ ॥
[57]मध्यपट्टं कलाश्रं चेच्छ्रीवज्र[58]स्तम्भमुच्यते ।
[59]अग्राकारं युगाश्रं स्यात् [60]त्रिपट्टक्षेपणान्वितम् ॥ २६ ॥
क्षेपणस्तम्भमित्युक्तं पट्टं पत्रादिशोभितम्[61] ।
ऊर्ध्वाधस्ताच्छिखामानं[62] त्रिचतुर्भागमेव वा ॥ २७ ॥
सर्वे पोतिकया[63] युक्ता[64] नानारूपैर्विचित्रिताः[65] ।

[दण्डलक्षणम्]

पादाग्रविपुलं यत्तु तद् दण्डमिति कथ्यते ॥ २८ ॥
[66]सर्वाण्यङ्गानि हर्म्याणां मानयेद्[67] दण्डमानतः ।

48. D : मुत्तो-
49. D : चतुरश्रसमं स्मृतम्
50. D : तदर्धे स्वर्धदण्डेन
51. D : शोडशाश्रं; C : विकारेण
52. E : ततो
53. D : मध्यपत्रं
54. D : मूलशेषं
55. D : मध्ये पट्टं
56. F : इष्यते
57. E : मध्येपट्टं; F : मध्यपट्टि
58. E : वर्ण- for वज्र-
59. F : अश्राकारं
60. D : त्रिपट्टं; F : त्रिपट्टे
61. A, D, F : पत्राद्विशोभितम्; E : भद्रादिशोभितम्; B : पत्रैर्विशोभितम्
62. D : मात्रं for मानं
63. D, E : वधिकया
64. D : युक्तं
65. C : नानारूपविचित्रिताः; D : नानारूपविचित्रितम्
66. A : सर्वाण्यङ्गानि; E : तत्तदङ्गनि
67. D : मापयेद्

bears the name of its decoration. Correspondingly then, that decorated throughout its height with protomes of elephants and which is provided with bell capital and abacus is called 'pillar with protomes' (*śuṇḍapāda*), that decorated with pearls throughout its height is called *piṇḍipāda*.

22b-26a A pillar of the *citrakhaṇḍa* type has a square element at the top two modules[14] high beneath which is an octagonal element with a doucine-like profile half a module high; beneath that again is a sixteen sided element one module high and next a (second) doucine profile, as above, then a square section median band one module high, then a (third) doucine profile and a (second) sixteen sided element as above; at the bottom, the remaining part of the height is occupied by a square element.[15] If the median band is octagonal section then the pillar is called *śrīkhaṇḍa* and if it is sixteen sided the pillar is called *śrīvajra*.

26b-27a If the upper part of the pillar is square section and if there is a projecting triple band the pillar is known as a *kṣepaṇastambha* and its triple band is decorated with leaves etc. . . .

27b At the summit and at the bottom of the pillar is a tenon a third or a quarter as wide (as the pillar).

28a All pillars are provided with a bracket capital and ornamented with various images.[16]

Definition of the module

28b-29a The module is the width of the top of the pillar and it is according to this module that all the measurements of every element of a building are to be taken.[17]

14. *Karṇāyāmena*: 'width two (standard) measures'; for the 'module' see below vv. 28b-29a. .
15. That makes up a total of eight and half modules, which agrees with proportions given above (v. 7).
16. *Potikā*: see below vv. 39b sq.
17. See 5.8 for the different *daṇḍa* used in measurements.

[कलशलक्षणम्]

सपादं सार्धपादोनद्विगुणं द्विगुणोन्नतम् ॥ २९ ॥
श्रीकरं चन्द्रकान्तं[68] च सौमुख्यं प्रियदर्शनम् ।
यथाक्रमेण नाम्नानि कलशानां[69] भवन्ति हि ॥ ३० ॥
[70]पोतिकाखण्डमण्डीनि कुम्भं स्कन्धं[71] च पद्मकम् ।
[72]मालास्थानं क्रमेणैव स्तम्भाग्रात्[73] परिकल्पयेत् ॥ ३१ ॥

68. D, F : रतिकान्तं
69. D : कलशानि
70. A, B, C : पोतिकाखण्डमध्यं च; D : बोधिकाखण्डिमण्डि च; E : बोधिकाखण्डमण्डञ्च; F : बोधिकाखण्डिमण्डि च
71. E : स्तम्भं
72. D : मालास्तानं; E : जलस्थानं
73. F : स्तम्भाग्रे

Bell capital

29b-30 Herewith, in order, the names of the (various) bell capitals (with their heights): *śrīkara* is one and a quarter (modules) high, *candrakānta* one and a half, *saumukhya* one and three quarters and *priyadarśana* one-half.[18]

31 Starting at the top of the pillar, there are successively: the bracket capital, the dye (featuring a human figure), the abacus, the bell capital, the support, the lotus and the band ornamented with garlands (*mālāsthāna*).[19]

18. The 'bell capital' is designated by terms whose first meaning is 'pot, vase' (*kumbha*, *kalaśa*) and which are applied to its main part (see below vv. 32 sq. and note 20). It is only a part of the crowning of pillar and its role is mainly decorative (like that of the abacus) as the main structural function is given to the *potikā* (below note 23). For its composition see below vv. 32-34a. According to *Ajita* (14.41-42) *śrīkara* bell capital is for circular or sixteen sided section pillars, *candrakānta* for octagonal ones, *saumukhya* for square one and *priyadarśana* for multifaceted ones (*atidhāra*).

19. Cf. *Īśānaśiva* (Kriyā. 31.60-61):

पोतिकाखण्डमण्डीनि कुम्भं लशुनमूलकम्।
अम्भोजमालास्थानानि स्तम्भाग्रे योजयेत् क्रमात्॥
अग्रहीनं तु लशुनं मूलोच्चं विपुलं भवेत्॥

and *Śilparatna* 21.116-17:

बोधिकाखण्डमण्डीनि कुम्भं लशुनमूलकम्।
अम्भोजमालां चानानि(?) पादपीठं पुनः क्रमात्
क्रमेण योजयेदग्रात् सर्वस्तम्भेषु बुद्धिमान्।
तत्र बोधिकया हीनं कारयेद् पादपीठकम्॥

For *potikā (bodhikā)*, *maṇḍi* and *kumbha*, see following notes. The *khaṇḍa* is most probably the element called elsewhere *vīrakaṇḍa* or *vīrakānta* (see v. 37 or 21.30, 33, 76, 82): it is a dye decorated with a human figure and which makes a recess between bracket capital and abacus. The support (*skandha* 'shoulder') is named in *Īśānasiva* 'garlic-bulb' (*laśuna*) which may indicate that it is wider at the base than at the top; it connects the bell-capital, above, and the 'lotus' , below. That last corresponds to the 'padmabandham' (according to Jouveau-Dubreuli terminology) and is a double row of lotus petals.

कुम्भोच्छ्रये नवांशे धृग्भागेन चतुरंशकैः ।
[74]कमलं कण्ठमंशेन भागेनास्यं प्रकीर्तितम्[75] ॥ ३२ ॥
भागेन पद्ममर्धेन[76] वृत्तमर्धेन हीरकौ[77] ।
[78]हीरौ पादसमव्यासौ[79] तत्कर्णेनास्यविस्तृतम्[80] ॥ ३३ ॥
तत्कर्णं कुम्भविस्तारं तत्कर्णं फलकायतम् ।
अथवा फलकायामं चतुर्दण्डं त्रिदण्डकम् ॥ ३४ ॥
सार्धत्रिदण्डमायाममुत्सेधाख्यं[81] त्रिदण्डकम्[82] ।
[83]तदुत्सेधे त्रिभागे तु भागेनोत्सन्धिरिष्यते ॥ ३५ ॥
भागेन वेत्रमंशेन पद्मं पाल्याभमिष्यते[84] ।
[85]नागवक्त्रसमाकारमावेत्रात्[86] पादरूपवत् ॥ ३६ ॥
पादविस्तारविस्तारं धृक्कण्ठं वीरकाण्डकम्[87] ।
सर्वेषामपि पादानां वीरकाण्डं[88] युगाश्रकम् ॥ ३७ ॥
तदुत्सेधत्रिपादोनं दण्डोत्थं[89] स्कन्धमिष्यते ।
तदधस्तु तदर्धेन पद्मं पत्रविचित्रितम्[90] ॥ ३८ ॥
मालास्थानमधस्तस्माद् दण्डमानसमुन्नतिः[91] ।

[पोतिका]

पादविस्तारविस्तारा[92] पोतिका[93] तत्समोदया[94] ॥ ३९ ॥

74. D : कलशं
75. D : प्रकल्पयेत्
76. D : पद्ममध्येन; E : पद्ममंशेन
77. D : हीनकौ; E : हीनकैः; F : भिरकौ
78. D and E : हीनौ; F : हीरौ
79. D : पादसमं व्यासौ
80. C : तत् कर्णोनास्यविस्तृतम्
81. D : उत्सेधं स्यात्; E : उत्सेधयास्यं
82. D : त्रिपादतः
83. D : तदुत्सेधं
84. D : पद्मवालुपमिष्यते
85. D : नागवृत्तसमाकारं वेत्रात् पादोनरूपवत्
86. E : वेत्रं
87. D : वीरकण्ढकम्
88. D : वीरकण्ढं
89. A : दण्डोद्धं; C : दण्डोऽर्धं; D, F : दण्डोच्चं
90. D, F : पद्मविचित्रितम्
91. D : दण्डमानं समुन्नतिः
92. D : विस्तारात्
93. D, E : बोधिका
94. C : तत्समोदयम्

32-34a The height of the bell capital[20] being divided into nine parts, the astragal takes up one part, the bell four, a groove one, a band with masks one, a doucine one, a reed half a one and the double *hīra* (?) half a one. The diameter of the double *hīra* is the same as that of the pillar and that of the band with masks is double that; the diameter of the bell is double (that of the pillar).

34b-39a The diameter of the abacus[21] is double (that of the pillar) or can be four, three and a half or three modules.[22] (The abacus) is three modules high and its height is divided into three (equal) parts: the upper connecting element occupies one, as do the reed and the inverted doucine; this abacus is in the form of a serpent's jaws; from the reed (downwards) its form is that of the pillar. The dye featuring a human figure is a support (?), its width is the same as that of the pillar and it is always square section no matter what the pillar. The support is a quarter the height of the dye (or may be) one module high; the doucine below it is half that height and the band ornamented with garlands is one module high and one module (wide) (?).

20. Descriptions of bell-capital given in the various treatises are most often muddled as appears in the following table where we have summarized those found in *Mayamata*, *Ajita* (14.34-36), *Dīptāgama* (5.23-24), *Īśānaśiva* (Kriyā. 31.51-53) and *Śilparatna* (21.79-80):

Mayamata	*Ajita*	*Dīptāgama*	*Īśānaśiva*	*Śilparatna*
dṛg 1/9	dṛg 1/9	dhṛg 1/9	dhṛg 1/9	dhṛg 1/9
kamala 4/9	kumbha 4/9	kalaśa 4/9	kumbha 4/9	ghaṭa 4/9
kaṇṭha 1/9	kaṇṭha 1/9	kaṇṭha 1/9	kaṇṭha 1/9	karṇa 1/9
āsya 1/9	āsya 1/9	āsya 1/9	mukha 1/9	mukha 1/9
padma 1/9	padma 1/9	padma 1/9	padma 1/9	padma 1/9
vṛtta 1/18	vṛtta-	vṛtta-	vṛtta 1/18	pīṭhikā 1/9
hīrakau 1/18	grīva 1/9	bhinnau 1/9	bhinnakau 1/18	

For a detailed commentary see Dagens 1984 p. 57.

21. *Maṇḍi:* see Dagens 1984 p. 58.

22. That is to say that the diameter of the abacus is two, three, and half or four times that of the pillar; cf. *Īśānaśiva* (Kriyā. 31.55b-56):

स्तम्भव्यासात् त्रिगुणितं चतुर्गुणमथापि वा ।
मण्ड्यायामः स्मृतोऽङ्घ्रयग्रसमो वेष्वब्धिभागतः ॥
मण्ड्युत्सेधो भवेत् तावान् वीरकाण्डस्य चोच्छ्रयः ॥

[95]पञ्चदण्डसमायामा श्रेष्ठार्धोच्चा[96] कनिष्ठिका ।
[97]आयता सा त्रिदण्डेन चतुर्दण्डेन दीर्घका[98] ॥ ४० ॥
[99]त्रिभागोना त्रिपादोच्चा मध्यमा पोतिका[100] भवेत् ।
पूर्वोक्तं तत् समण्डीनां सकुम्भानां[101] चतुर्गुणम् ॥ ४१ ॥
[102]त्रिगुणं केवलानां तु पादानां प्रविधीयते ।
सर्वेषामपि पादानां[103] यथेष्टायतमीरितम् ॥ ४२ ॥
तदुच्चत्रिचतुर्भागोच्चा वा स्वाग्रे तु पट्टिका ।
अर्धं त्रिद्व्यंशमङ्घ्र्यूनं छायामानं[104] विधीयते ॥ ४३ ॥
त्रिभागं वा चतुर्भागं तरङ्गस्थानमिष्यते ।
सक्षुद्रक्षेपणं मध्यपट्टं[105] पत्रविचित्रितम् ॥ ४४ ॥
[106]समास्तरङ्गा[107]श्चान्योन्यहीनाः सर्वत्र सम्मताः ।
अग्रनिष्क्राममर्धं वा त्रिभागं वा स्वतारतः ॥ ४५ ॥
मुष्टिबन्धोपरिक्षिप्त[108] व्यालसंहतिरूपवत्[109] ।
सनालीकं समतलं सनाटकमथापि[110] वा ॥ ४६ ॥
[111]भूतेभमकरैर्व्यालसंयुक्तं चाग्रमण्डनम् ।
[112]पार्श्वयोः पोतिकामध्ये पट्टं पादविशालवत्[113] ॥ ४७ ॥

95. D : replaces 40-41 by:
पञ्चदण्डसमायामं श्रेष्ठं बोधिका भवेत् ।
आयतं स्यात् त्रिदण्डेन मध्योच्चकनिष्ठिका ॥
चतुर्दण्डेन दैर्घ्यं स्यात् विस्तारं पादपत्तति ।
त्रिभागेन द्विभागोच्चं मध्यया बोधिका भवेत् ॥
पूर्वोक्तं तत्समण्डीनां कुम्भानां च चतुर्गुणम् ।
फलकेन समायुक्तं चतुर्दण्डेन दैर्घ्यकम् ॥
त्रिभागोन त्रिभागोच्चं मध्यमा बधिका भवेत् ।
96. E : श्रेष्ठोर्ध्वं च
97. F : omits 40b-47b.
98. A : दिर्यका; C : दीर्घतः; E : दैर्घिका
99. E : त्रिभागोन त्रिभागोच्चा
100. E : बोधिका
101. A : सकुम्भा-
102. E : त्रिकोणं
103. D : देवानां
104. A : अङ्घ्ररुनं छायामानं; C : अनोन छायामानं
105. E : मध्यवर्गं
106. D : omits 45a.
107. A, C : तुरङ्गाश्च-
108. D : क्षिप्य for क्षिप्त-
109. D, E : व्यालसिंहादिरूपवत्
110. C : सनागकमथापि
111. D : भूतैर्मकरैर्व्यालैः संयुक्तं
112. D : द्विपार्श्वे बधिकामध्ये
113. All texts with the exception of D, place 47b between 46a and 46b.

Bracket capital[23]

39b-42 The bracket capital is the width of the pillar and is as high as wide; the largest is five modules long and half that high, the smallest three module long and the medium four module long with a height five twelth that (?). These measurements multiplied by four give (the total height) of pillars comprising an abacus and a bell capital whilst, multiplied by three, they give the height of pillars lacking these elements. The selected height should be used for all the pillars (on any one storey of a building).

43-47 The top of the bracket capital[24] is a string-course a third or a quarter its height and whose overhang[25] is half, two-thirds or three quarters (its width). The height of that part of the bracket capital which is decorated with waves is a third or a quarter (that of the bracket capital); this element is provided with a median band, slightly projecting and decorated with foliage; the height of the waves is invariably constant but they grow progressively narrower (starting from the centre). (The element which is decorated with the waves) has, at its ends, a projection, half or one-third of its width, which is in the form of the tail of a *makara* and surmounts a fist-like decoration;[26] it may have (at both ends), a gargoyle or a *matala* (?) or a dancing figure and, at the top, a decoration of dwarves, elephants, *makara* or *vyāla*. The median band of the bracket capital is the same width on both sides of the pillar.

23. 'Bracket-capital' is a poor substitute for rendering *potikā* (or *bodhikā*); this elongated piece placed between the pillar and the entablature is that we may call a 'support for the entablature' as proposed by P. Stern (in *Colonnes indiennes d'Ajanta et d'Ellora,* 1972, p. 25). For detailed account on that element see Dagens 1984 p. 59-61.

24. *Śilparatna* (21.102-04) gives a slightly clearer description:
वोधिकोच्चे त्रिभागे तु भूम्यंशेनोर्ध्वपट्टिका ।
मध्येऽंशेन तरङ्गः स्याल्लताद्यैर्भूषितं तु वा ॥
तदधस्त्वेकभागेन मुष्टिबन्धं तु कारयेत् ।
त्रिभागे वोधिकोत्सेधे सार्धांशेनाग्रपट्टिका ॥
तदर्धार्धेन तदधः क्षेपच्छायान्वितं भवेत् ।
तदुच्चार्धात् त्रिभागानामंशाभ्यां वात्र कल्पयेत् ॥

25. *Chāyāmāna*: 'the measure of the shadow';

26. *Muṣṭibandha*: these decorative elements shaped like balls or 'fists' (hence their name), are hanging from *potikā* ends or may be applied on a string-course or on the ends of the beams (see 16.10).

[114]रत्नबन्धक्रियावल्ली चित्रा [115]वाग्रस्थपट्टिका ।
[116]नानाचित्रैर्विचित्रा वा सा प्रोक्ता चित्रपोतिका[117] ॥ ४८ ॥
पत्रैर्विचित्रिता पत्रपोतिकेति[118] प्रकीर्तिता ।
महार्णवतरङ्गाभतरङ्गाभा[119] तरङ्गिणी ॥ ४९ ॥
[120]चतुःषडष्टपङ्क्त्यर्कसंख्या वा स्युस्तरङ्गकाः ।
बहवोऽपि समाश्चैते चान्योन्याः स्युर्वराः[121] क्रमात् ॥ ५० ॥

[स्तम्भविषये विशेषाः]

पादमर्धं त्रिपादं वा भित्तेः स्तम्भस्य[122] निर्गतम् ।
चतुरष्टाश्रवृत्तानां यथाक्रममिति स्मृतम्[123] ॥ ५१ ॥
[124]द्विहस्ताद्यं चतुर्हस्तं स्तम्भान्तरमिति स्मृतम् ।
[125]षडङ्गुलविवृद्ध्या तु नवभेदं[126] प्रकीर्तितम् ॥ ५२ ॥
गृहीतांशवशेनापि यथायुक्त्या प्रयोजयेत्[127] ।
[128]स्तम्भस्तम्भान्तरं सर्वं[129] प्रासादे सार्वदेशिके ॥ ५३ ॥
[130]विषमस्तम्भभागं तु [131]वास्तुवस्तुविनाशनम् ।
सायतं चापि तत्सर्वं तत्राम्नैव[132] प्रपद्यते ॥ ५४ ॥
दारुस्तम्भविशालं वा सार्धं द्वित्रिगुणं तु वा[133] ।
शिलास्तम्भविशालं स्याद् देवानां नैव[134] मानवे ॥ ५५ ॥

114. A, B, C, D : रक्त-
115. A, B : वा ग्रन्थपट्टिका; D : वस्त्रग्रपट्टिका
116. D : नानाविचित्रचित्रं स्यात् तत् प्रोक्तं चित्रबोधिका
117. C : चित्रपट्टिका
118. D, E : बोधिकेति
119. E : तरङ्गाङ्ग-
120. D : चतुःषडष्टपङ्क्तर्क संख्यायाङ्गस्तरङ्गकाः
121. A, C : स्याद्वराः
122. D : तुङ्गस्य
123. D : उदीरितम् for इति स्मृतम्
124. A, C : द्विहस्ताद्या चतुर्हस्तात्
125. D : षट् षडङ्गुलवृद्धा वा
126. E : नवभागं
127. D : युक्त्यादि योजयेत्
128. D, E, F : समस्तम्भान्तरं
129. C : सम्यं
130. C : विषमं स्यादन्तरं तु
131. D, E : वस्तुवास्तु-
132. D : गणनन for तत्राम्नैव
133. E : द्विगुणतुङ्गतः
134. D : चैव

48-50 (The bracket capital) is called *citrapotikā* if decorated with foliage and (garlands) of jewels or if its upper string-course is decorated in various colours; it is called *patrapotikā* if decorated with foliage. If it is decorated with waves like those of the Ocean it is then called *taraṅgiṇi*. These waves number four, six, eight, ten, twelve or even more; they are equal (in height) and become progressively wider (from the sides inwards).

Further features of pillars

51 According to whether (its section) is square, octagonal or circular a pilaster's projection[27] will be a quarter, a half or three quarters its width.

52-54 The intercolumniation should be from two to four cubits which, if six digits are added each time, gives nine possibilities.[28] After a suitable intercolumniation has been chosen it is to be used for all the pillars of any one building no matter what its type.[29] If the disposition of the pillars is irregular this will bring destruction upon the building and upon its site but, if the whole arrangement is true to the principle, success will ensue.

55-56 The diameter of a stone pillar is the same as that of a wooden one or may be half as much again or double; such a pillar is appropriate for buildings intended for gods but not for those intended for men. According to the ancients, the pillars may be (of) brick, stone or wood

27. *Bhitteḥ stambhasya nirgamam*: or 'the projection of the pillar from the wall', however the corresponding passage in *Īśānaśiva* (Kriyā. 31.33b-34a) does not favour that interpretation:

पादमर्धं त्रिभागं च भित्तिस्तम्भविनिर्गमम् ।
चतुरश्राष्टवृत्तानां विज्ञेयस्तु क्रमाद् भवेत् ॥

28. For intercolumniation in pavilions see 25.13 sq.

29. *Prāsāde sārvadeśike* (see 15.59, 20.6, 24.77, etc.). This expression may refer to the so-called 'geographical' classification of building (*nāgara, drāviḍa, vesara*,... etc.); cf. 19.35 sq. and note *ad loc.*

[135]इष्टकाश्मद्रुमैः सर्वैः स्तम्भाः प्रोक्ताश्चिरन्तनैः ।
युग्मायुग्मं तु देवानामयुग्मं तु नृणां मतम्[136] ॥ ५६ ॥
अन्तःस्तम्भं बहिःस्तम्भमाजुसूत्रं यथा भवेत्[137] ।
[138]गृहाणां भित्तिमध्ये तु शालानां तु तथा भवेत् ॥ ५७ ॥
प्रासादानां तु पाद्वाह्ये पान्मध्ये शयनासने ।
[139]उपानादिशिरः[140] केचित् केचित् स्तूप्यन्तमुन्नतम् ॥ ५८ ॥
मुनयः प्रवदन्त्युच्चं प्रासादे सार्वदेशिके ।
पाद्वाह्ये पादमध्ये वा सभामण्डपयोर्मतम्[141] ॥ ५९ ॥
अन्तर्बहिश्च मध्ये तु सालानां मानसूत्रकम् ।
[142]युञ्जीयादेवमेवं तु सर्वेषां सम्पदां पदम् ॥ ६० ॥
विपरीते विपत्त्यै[143] स्यादिति शास्त्रविनिश्चयः[144] ।

[द्रव्यपरिग्रहः]

स्तम्भोत्तरादिकाङ्गानां द्रव्यं[145] द्रुमोपलेष्टकाः ॥ ६१ ॥

[वृक्षलक्षणम्]

स्निग्धसारमहासारा [146]ह्यवृद्धास्तरुणेतराः ।
[147]अवक्रा निर्व्रणाः सर्वे ग्रहीतव्या महीरुहाः ॥ ६२ ॥
[148]पुण्याद्रिवनतीर्थस्था दर्शनीया मनोरमाः ।
सर्वसम्पत्समृद्ध्यर्था[149] भवेयुस्ते न संशयः ॥ ६३ ॥

135. D : omits 56; F : omits 56-68a.
136. C : भवेत् for मतम्
137. E : यथा तथा
138. D : गृहामभिन्नमध्ये
139. D : उपनादिशिरः केचित् स्तूप्यन्तश्चमुदं विदुः
140. E : उपानाजिशिरः
141. E : स्मृतम् for मतम्
142. C : युञ्जीत देवदेवं
143. D : विपद्ये
144. D : शास्त्रस्य निश्चयम्
145. D : द्रव्यात्
146. D : वृद्धश्च for ह्यवृद्धास्
147. D : अकरा
148. D : पुण्यभूवनतीर्थं स्याद् दुर्गातिर्वा मनोरमा
149. D : समृद्धिस्स्यात्

but should be homogeneous[30] (in any one building) and their number, which may be either even or odd for gods, must be odd for men.

57-58a The measuring line[31] goes along the inside or the outside of the row of pillars (which surrounds the building) but, in houses or *śālā*, it is along the median axis and in temples exterior to their pillars and, in bedrooms, on their axis (?).

58b-59a According to some sages the height of a building of any type is measured from the plinth (of the base) to the roof but, according to others, it is measured from plinth to finial.[32]

59b-61a In assembly halls and pavilions the measuring line goes along the axis of pillars or along their exterior, but it should go along the interior, the exterior or the median axis of (enclosure) walls; whichever the case this is a source of good fortune but failure is certain if there is any deviation from the rules, as is affirmed by the treatises.

Choice of building materials[33]

61b Wood, stone and brick are the materials to be used for (vertical) elements such as pillars and (horizontal) elements such as architraves.

Trees

62-63 The chosen trees must be perfect, hard and vigorous; they should neither be old nor should they be saplings; they should not be crooked and should be undamaged; they should be growing in a holy place, mountain, wood or river-bank and they should be pleasing to eye and to mind. It is a certainty that such trees are conducive to prosperity and good fortune.

30. Literally: 'all (pillars) are in bricks, stones or wood'.
31. *Ājusūtra* (and below v. 60: *mānasūtra*): see 6.19b sq. and note *ad loc.* About the position of that line see also 21.21, 22.79-81, 23.17, 25.195 . . .
32. This verse seems to be out of place as it interrupts the statement on measuring line position.
33. On that subject also see 18.92 (coatings and mortars), 18.164 (woods suitable for making the finial axis), 25.182 (woods for pavilions), 33.4 sq. (stones for making Linga). cf. Dagens 1984 pp. 20-29.

पुरुषः खदिरः सालो मधूकः चम्पकस्तथा[150] ।
शिंशपार्जुनाजकर्णी क्षीरिणी[151] पद्मचन्दनौ[152] ॥ ६४ ॥
पिशितो धन्वनः पिण्डी सिंहो राजादनः शमी ।
[153]तिलकश्च द्रुमाश्चैते स्तम्भवृक्षाः समीरिताः ॥ ६५ ॥
[154]निम्बासनशिरीषाश्च एकः[155] कालश्च कट्फलः ।
[156]तिमिसो लिकुचश्चैव पनसः सप्तपर्णकः ॥ ६६ ॥
[157]भौमा चैव गवाक्षी चेत्यादयश्चोर्ध्वभूरुहाः ।

[शिलालक्षणम्]

[158]एकवर्णाः स्थिराः स्निग्धाः सुखसंस्पर्शनान्विताः ॥ ६७ ॥
प्राचीनाश्चाप्युदीचीना भूमग्नाः[159] शुभदाः शिलाः ।

[इष्टकालक्षणम्]

स्त्रीलिङ्गाश्चापि पुल्लिङ्गा निर्दोषाश्च नपुंसकाः ॥ ६८ ॥
[160]सुघनाः समदग्धाश्च [161]सुस्वराश्चेष्टकाः शुभाः ।
स्त्रीलिङ्गाश्चापि पुल्लिङ्गा भिन्नच्छिद्रादिवर्जिताः ॥ ६९ ॥
एतैरेवंविधैर्द्रव्यैः कृतं[162] वस्तु समृद्धये[163] ।
धर्मार्थकामसौख्यानां[164] भवेदेवेति निश्चितम्[165] ॥ ७० ॥

150. A, C : स्तम्बकस्तथा; B : स्तबकस्तथा
151. E : कण्टकी
152. C : पद्मचन्दनाः
153. D : किलकश्च
154. D : निम्वासनिश्चीर्षवाकादलकश्च कक्षत्फलाम्
155. E : शाकः
156. A, C : किमिसो
157. D : होमश्चैव; E : हेमश्चैव
158. D : एकवर्णासनास्निग्धा
159. D : भूमन्ताः
160. D : निश्शेषसुखनासं दग्धाश्चसुसुखश्चेषका [gap]-भाः
161. E : सुखना-; F : स्वस्वरा-
162. D : कृत- instead of कृतं
163. C : समृद्धिभत्
164. C : मोक्षानां instead of सौख्यानां
165. D and E : निश्चयः

64-67a Herewith the trees suitable for the shafts of pillars: *puruṣa*, *khadira*, *śāla*, *madhūka*, *campaka*, *śiṃśapa*, *arjuna*, *ajakarṇin*, *kṣīriṇī*, *padma*, *candana*, *piśita*, *dhanvana*, *piṇḍī*, *siṃha*, *rājādana*, *śamī* and *tilaka*. Equally suitable are *nimba*, *āsana*, *śirīṣa*, *eka*, *kāla*, *kaṭphala*, *timisa*, *likuca*, *panasa*, *saptaparṇaka*, *bhaumā* (?) and *gavākṣīn*.

Stone[34]

67b-68a Suitable stones are of even colour and are hard, perfect, pleasing to the touch and are embedded in the earth with an eastward or northward orientation.

Bricks

68b-70 Bricks are female, male or neuter[35] and must be without flaw, compact and uniformly baked; they should, as well, give off an harmonious sound. Male, female and neuter bricks must be free from fissures (?) and cracks. It is well-known that a building constructed out of such materials is conducive to the growth of Dharma, Artha and Kāma.

34. For more detailed accounts see 33.4 sq. or *Śilparatna* 14.1.44.
35. See 12.105-6.

[वर्ज्या वृक्षाः]

न देवतालयान्तस्थाः प्रहता वा न विद्युता[166] ।
[167]न दावानलसंलीढा न भूतालयमध्यगाः ॥ ७१ ॥
न महापथ[168] संरूढा न तु ग्रामसमुद्भवाः ।
[169]न घटाम्बुभिरासिक्ता न पक्षिमृगसेविताः ॥ ७२ ॥
न वायुना न मातङ्गैर्भग्ना नैव गतासवः[170] ।
[171]न चण्डालजनाकीर्णा न सर्वजनसेविताः ॥ ७३ ॥
नान्योन्यवलिता[172] भग्ना न वल्मीकसमाश्रिताः ।
[173]न लतालिङ्गिता गाढा न सिराकोटरावृताः ॥ ७४ ॥
नाङ्कुरावृतसर्वाङ्गा न भृङ्गकीटदूषिताः[174] ।
[175]नाकालफलिनो ग्राह्या न श्मशानसमीपगाः[176] ॥ ७५ ॥
[177]सभाचैत्यसमीपस्था देवादीनां न भूरुहाः ।
[178]वापीकूपतटाकादिवस्तुष्वपि[179] च सम्भवाः[180] ॥ ७६ ॥
[181]विनष्टवस्तुसञ्जातद्रव्यं सर्वविपत्करम् ।
तस्मात् सर्वप्रयत्नेन शुद्धं द्रव्यं[182] प्रगृह्यताम् ॥ ७७ ॥
शिला देवालये ग्राह्या द्विजावनिपयोर्मताः[183] ।
[184]पाषण्डिनां च कर्तव्या न कुर्याद् वैश्यशूद्रयोः ॥ ७८ ॥

166. D : प्रभुता वा विद्युता
167. D : वनदावानलसंलीढा
168. D : पद्म- for पथ-
169. D : त न for न
170. C : स्वभावतः
171. D : चण्डालवाजनाकीर्णो
172. D : फलिता for वलिता
173. E : न वीनलिङ्गता
174. A, B, E, F : भृशं कीटदूषिताः; C : भृषं कीटभूषिताः
175. D : न कालफलिनो
176. D : समीपके
177. F : omits 76a.
178. D : omits 76b.
179. E : वास्तुष्वपि
180. C : भभवाः
181. D : विनिष्टवस्तु यत् पूर्वं शल्यं सर्वविपत्करम्
182. C and D : शुद्धद्रव्यं
183. D : द्विजा वा नृपयेर्मताः; F : द्विजावनिचयोकृतः
184. D, E, F : तापसानां

Trees to be avoided

71-76 The (chosen) trees should not be close to temple and should not have been struck by lightning nor scorched by any conflagration; they should not be growing in a place inhabited by spirits nor beside a major roadway nor in a village and they should not have been sprinkled with the water from the ritual (and thus be objects of cult worship); they should not be frequented by birds nor by wild animals nor should they have been bent by wind nor by elephants; they should not be dead nor be serving as shelter for caṇḍāla nor for men of any caste whatsoever; they should not be entangled with each other nor should they be twisted; they should not harbour white ants nor be strangled by liana nor be bound; they should not be hollow nor have empty veins (?), there should be no twigs whatsoever on their branches; they must not have been damaged by wild bees nor by worms nor should they bear fruit out of season; they should not be propped up and should not grow near a cemetery nor assembly hall nor holy ground; they must not belong to any god or deity nor should they be found near a tank, well, pond or like place.

77 Materials from a forbidden locality give rise to all sorts of misfortune and the correct materials should therefore be chosen with the utmost care.

78-79a Stones should be used for temples and for the dwellings of brahmins and of the king as well as for heterodox (shrines) but they are

कर्तव्यं यदि तद् वास्तु[185] धर्मकामार्थनाशकृत्[186] ।
एकद्रव्यकृतं शुद्धं मिश्रं द्विद्रव्यनिर्मितम्[187] ॥ ७९ ॥
[188]त्रिद्रव्यसंयुतं यत्तु तत् सङ्कीर्णमुदाहृतम् ।
[189]पूर्वोदितानां वासेषु कर्तव्यं सम्पदां पदम् ॥ ८० ॥

[वृक्षसंग्रहणम्]

सर्वद्वारिकनक्षत्रे शुभपक्षमुहूर्तके ।
गच्छेदरण्यं द्रव्यार्थी[190] कृतकौतुकमङ्गलः ॥ ८१ ॥
निमित्तैः शकुनैर्योग्यैः सह[191] मङ्गलशब्दकैः ।
गन्धैः पुष्पैश्च धूपैश्च[192] मांसेन कृसरेण च[193] ॥ ८२ ॥
पायसौदनमत्स्यैश्च भक्षैश्चापि[194] पृथग्विधैः ।
अर्चयेदीप्सितान् सर्वान् वृक्षांश्च वनदेवताः॥ ८३ ॥
भूतक्रूरबलिं दत्त्वा कर्मयोग्यद्रुमं हरेत्[195] ।
मूलाग्रादार्जवं वृत्तं[196] शाखानेकसमन्वितम् ॥ ८४ ॥
[197]तत्तु पुंस्त्वं भवेन्मूले स्थूलं स्त्रीत्वं कृशाग्रकम् ।
स्थूलाग्रं कृशमूलं तु षण्डमेतदुदीरितम्[198] ॥ ८५ ॥
मुहूर्तस्तम्भमुद्दिश्य पुंभूरुह उदीरितः[199] ।
[200]सर्वेष्वङ्गेषु वस्तूनां[201] पुंस्त्रीषण्डं प्रकीर्तितम् ॥ ८६ ॥
पूर्वाशायां द्रुमस्यास्य स्वपेद्[202] दर्भान्तरे शुचिः ।
[203]स्वप्रदक्षिणपार्श्वे तु संस्थाप्य परशुं सुधीः ॥ ८७ ॥

185. D, E : मोहेन instead of तद् वस्तु
186. F : नाशनम्
187. D : द्विद्रव्यसंयुतम्
188. D : निर्मितत्रसंयुक्तं संकीर्णं तत् प्रकीर्तितम्
189. D : पूर्वोदितानि सर्वेषु
190. D : द्रव्यार्थं; E : द्रव्यानां
191. D : सर्व-
192. F : मत्स्यैश्च instead of धूपैश्च
193. D : मांसैश्च रुधिरेण वा
194. D : हविश्चापि instead of भक्षैश्चापि
195. A, C, E, F : वरैः ; D : खनेत्
196. E : विप्रं; F : कृत्वा
197. D : तत् पुंस्त्वं भवेत् स्त्रीत्वं मूलस्थूलं कुशाग्रकम्
198. D, E : उदाहृतम्
199. D, F : उदाहृतम्
200. D : सर्वेष्वङ्गेषु या पुंसि पुंस्त्रीषण्डा प्रकीर्तिताः
201. E : वास्तूनां
202. E : श्वभ्रे for स्वपेद्
203. D, E, F : स्वप्रदक्षिणपार्श्वे तु परशुं ग्राह्य बुद्धिमान् ।

inappropriate for vaiśya and śūdra and such a building so constructed will bring about the disappearance of Dharma, Artha and Kāma.

79b-80 A building constructed out of only one material is said to be 'pure' whilst with two materials it is called 'mixed' and with three, 'mingled'. Success emanates from houses built according to the above mentioned principles.

Search for trees[36]

81-82a One who wishes to look for materials should go into a forest, under a constellation favourable to the enterprise and during a bright fortnight; he must previously have accomplished the solemn preliminary rites and it is essential that the portents and omens be auspicious and that sounds of good omen accompany him (on his expedition).[37]

82b-83 To begin with, the chosen trees are to be propitiated, as also the forest deities, with the aid of perfumes, flowers and fumigations and with offerings of meat, boiled rice with sesame, rice with water and fish and with food of all kinds.

84-86 Once bloody offerings have been made to the Spirits, the tree suitable for the (projected) work should be selected. A tree which is straight, cylindrical from root to top and which has plenty of branches is called 'male' whereas it is 'female' when broad at the bottom and thin at the top, and neuter when it is broad at the top and thin at the bottom. A tree intended for a *muhūrtastambha*[38] should be male, but male, female and neuter trees are suitable for every part of a building.

87-88 The architect, purified and wise, is to spend the night on a spread of *darbha* to the east of the tree having first placed his axe beside him to the right. Then in the morning, after drinking pure water and facing to

36. The ceremony described here is very similar to that prescribed for the making of the pole for the god's banner (*dhvajadaṇḍa*); cf. e.g. *Rauravāgama* Kriyā. 18.22 sq.

37. The list of good and bad omens is given in *Ajita* (5.2-14).

38. On that pillar placed above the foundation-deposit see below vv. 104b sq. and 27.66 sq.

पीत्वा शुद्धं पयो रात्रावपराभिमुखोऽपरः[204] ।
स्थपतिर्वरवेषाढ्यो मन्त्रयेत् सपरश्वधः[205] ॥ ८८ ॥
अयं मन्त्रः —
अपक्रामन्तु भूतानि देवताश्च सगुह्यकाः[206] ।
[207]युष्मभ्यं तु बलं भूयः[208] सोमो दिशतु पादपाः! ॥ ८९ ॥
शिवमस्तु महीपुत्रा ! देवताश्च सगुह्यकाः ! ।
[209]कर्मैतत् साधयिष्यामि क्रियतां वासपर्ययः ॥ ९० ॥
[210]एवमुक्त्वा नमस्कृत्य पादपेभ्यो नमः शुचिः ।
[211]दुग्धतैलघृतैः सम्यक् सन्तेज्य[212] परशोर्मुखम्[213] ॥ ९१ ॥
उपक्रामेत्तु तं छेत्तुं यथाकामं वनस्पतिम् ।
[214]मूले हस्तं व्यपोह्योर्ध्वे त्रिश्छित्वा तत्र लक्षयेत्[215] ॥ ९२ ॥
वारिस्रावो विवृद्ध्यर्थः क्षीरं पुत्रविवर्धनम्[216] ।
[217]शोणितं स्वामिनं हन्याद् वर्जयेत् तं प्रयत्नतः ॥ ९३ ॥
पतने सिंहशार्दूलहस्तिशब्दाः सुशोभनाः[218] ।
[219]रुदितं हसितं कोशं[220] कूजितं निन्दितं वरैः[221] ॥ ९४ ॥
[222]पातयेदुत्तराग्रं तु पूर्वाग्रं वा वनस्पतिम् ।
ते दिशौ शुभदे[223] स्यातामन्याशासु[224] विपर्यये ॥ ९५ ॥

204. E : वरः for [अ]परः
205. A, B, C, D : सपरश्वथः
206. D : सरक्षसान्
207. E : omits 89b-90a.
208. D : बलिं भूयात्
209. D : किमेतत्
210. D : एवमुक्तान्
211. F : द्रव्य-
212. C : संमृज्य; D : संत्यज्य; E : संशोद्ध्य
213. E : परशोर्भुवम्
214. D : मूले हस्ते; E : मूलहस्तं
215. E : लक्षणम्
216. D : पुत्रवर्धनम्
217. D : शोणितं स्वामिनाशं स्याद्
218. D : शब्दैस्तु शोभनम्
219. F : रुद्रकं हस्तकं
220. C : क्रोशः
221. D : रुदितं परैः
222. D : पतने
223. D : शुभदौ
224. C : अन्यास्तु स्युर्

the west, the perfect architect, clad in white and with axe in hand, pronounces this formula:

89-90 Herewith the formula: "Let Spirits, Divinities and Demons disperse! On you O trees let Soma bestow power! May It be propitious for you O sons of Earth! Divinities and Demons I shall accomplish this act and you must change your dwelling place."

91-93 The pure architect, having done this and having paid homage to the tree, whets the edge of the axe suitably with milk, oil and ghee[39] and then embarks upon the cutting down of the selected tree. Having left one cubit at the base, he knocks three times and begins his examination; if water flows out this will bring about an increase in fortune and if milk-white sap (the house owner) will have many sons but if there is blood-red sap this will mean the death of the owner (of the house) and the greatest care should be taken to avoid such a tree.

94 If the cries of lions, tigers or elephants are heard at the moment when the tree falls this is a good omen but the sages condemn tears, laughter, shrieks and whispers (heard at that moment).

95-98 It is auspicious that the tree should fall to the north or the east and it augurs badly if the fall is to any other direction. It is lucky if it falls onto *sāla*, *aśmari* or *ajakarṇin* trees but if the tree falls upside down[40] this will cause the death of the relatives and servants (of the householder).

39. According to *Īśānaśiva* (Kriyā. 33.24) this is done with *pañcagavya*.
40. Literally: 'if the root falls above the top'.

[225]सालाश्मर्यजकर्णीनामूर्ध्वे तु पतनं[226] शुभम् ।
[227]मूले पृष्ठागमे बन्धुप्रेष्ययोश्च विनाशनम् ॥ ९६ ॥
[228]निर्गमत्स्थितिमद् भूत्वा वृक्षान्तरनिपातने ।
शिरःसङ्गेन नाशः[229] स्यान्मूल[230] सङ्गे श्रमो भवेत् ॥ ९७ ॥
शरीरभङ्गं कर्तॄणां नाशमग्रेऽप्यपत्यहृत्[231] ।
[232]अन्योन्यपतनं पूज्यं छेद्यं चोभयतः समम् ॥ ९८ ॥
[233]चतुरश्रमृजुं कृत्वा मुहूर्तस्तम्भसंग्रहे । ।
सितपट्टेन सञ्छाद्य स्यन्दने न्यस्य वेशयेत्[234] ॥ ९९ ॥
देवद्विजमहीपानां विशां वै शकटेन तु ।
शूद्रस्य पुरुषस्कन्धेनानीयात्तु विचक्षणः ॥ १०० ॥
पार्श्वयोः शाययित्वा[235] तु शकटे न्यस्य[236] वेशयेत् ।
प्रशस्ते द्वारि प्रग्राह्य स्थपत्यनुगतद्रुमम्[237] ॥ १०१ ॥
[238]कर्ममण्डपके न्यस्य बालुकोपरि शाययेत्[239] ।
[240]प्रागग्रं चोत्तराग्रं वाप्याशुष्कं रक्षयेत् पुनः ॥ १०२ ॥
परावृत्तं न कर्तव्यमाषण्मासं तु स द्रुमः[241] ।
[242]सर्वेन्द्रकीला एवं स्युः[243] प्रापणीयाः[244] प्रयत्नतः ॥ १०३ ॥
[245]अन्येषामपि कुप्यानां[246] वेशने त्वग्रमग्रतः ।

225. D : शालाकर्याजकर्णीनां
226. C : ऊर्ध्वाग्रपतनं
227. D : adds between 96a and 96b. सालमत्स्वा-जकर्णं पनसं चम्पकं तथा ।
228. D : निर्गम्य स्थितिं भूत्वा वृषान्तरे तु पातने
229. D : नाशं 230. D : मूलं
231. D : त्वपत्युनम्
232. D : अन्ये तत् पतनं
233. D : ऋजुकृत्वाग्रचतुश्रमुहूर्तस्तम्भसङ्ग्रहे ।
234. E : स्यन्दनैः स्थाप्य वेशनम्
235. E : साधयित्वा
236. D : शकटेनास्य
237. D : स्थपत्यानुगतद्रुमम्
238. D : धर्ममण्डपके
239. D : शाधयेत्; E : धापयेत्
240. D : प्राक् शिरश्चोत्तराग्रं वा शुष्कं रक्षयेत् ततः
241. D : तद् द्रुमम्
242. All texts place 103b-104a after v.100.
243. D : सर्वेषामपि कीला स्युः
244. E : प्रापणीयात्
245. D : एवमन्येऽपि क्रय्यानां वेशयेत् पूर्वमग्रतः ।
246. F : रूपानां

If, by falling amongst surrounding trees, the tree remains in place once cut this presages death, if it is held back at the top, and illness, if it is held back at the bottom. If the trunk of the tree is broken this heralds the death of the wood-cutter and if its top is broken this will mean the disappearance of his line; preparation must thus be made for the fall of the tree, at its top and its base (?).

99-100 The tree is to be cut at both its extremities and, if a *muhūrtastambha* is wanted, it is squared off then covered with white cloth and hoisted onto a litter. The well advised man will, however, have it transported by chariot if it is intended for a god, a brahmin, a king or a vaiśya though for a śūdra he will have it carried on a man's back.

101-102 The tree is laid on its side and then hoisted into the chariot. Once the tree selected by the architect has been taken into the workshop through the door prescribed, it is installed by being drawn up onto the sand with its top towards the east or the north.[41] The tree is to be kept without being moved for six months, until it is dry.[42]

103-104a In the same way all the nails should be stowed away with great care along with the other metal pieces and all these things should point in the same direction (?).[43]

41. Thus the orientation of the tree in the workshop is to be the same than that of its fall in the forest (cf. v. 95).
42. According to *Īśānaśiva* (Krīya. 33.34) the tree is to be kept for three, five or eight months. For the sake of comparison we may add that, nowadays in Europe, hard woods are to be kept for one year by centimetre of thickness and soft woods for six months.
43. Cf. *Mānasāra* 15.340.

[मुहूर्तस्तम्भः]

मुहूर्तस्तम्भो देवानां द्विजातीनां [247]यथाक्रमम् ॥ १०४ ॥
[248]कार्तमालश्च खदिरः खादिरश्च मधूककः ।
[249]राजादनो यथासंख्य विस्तारायाममुच्यते ॥ १०५ ॥ ।
भानुरुद्रदशद्वारवितस्त्यायामसंयुताः[250] ।
तत्संख्याङ्गुलिविस्तीर्णाः[251] पङ्क्त्यंशोनाग्रविस्तराः ॥ १०६ ॥
भूतसार्धचतुर्वेदगुणतालनिखातकाः[252] ।
भूमिभूमिवशादुक्तं[253] स्तम्भोच्चं विपुलं तु वा[254] ॥ १०७ ॥
झषालाङ्घ्रौ तु सर्वत्र निखातं परिवर्जयेत् ।
अश्वत्थोदुम्बरश्चैव प्लक्षश्च वटवृक्षकः ॥ १०८ ॥
सप्तपर्णश्च बिल्वश्च पलाशः कुटजस्तथा[255] ।
[256]पीलुः श्लेष्मातकी लोध्रः कदम्बः पारिजातकः ॥ १०९ ॥
शिरीषः कोविदारश्च तिन्त्रिणीको महाद्रुमः ।
शिलीन्ध्रः सर्पमारश्च शल्मली सरलस्तथा[257] ॥ ११० ॥
किंशुकश्चारिमेदश्चाभयाक्षा[258] मलकद्रुमाः ।
कपित्थः कण्टकश्चैव[259] पुत्रजीवश्च डुण्डुकः ॥ १११ ॥
[260]कारस्करः करञ्जश्च [261]वरणश्चाश्वमारकः ।
बदरो वकुलः पिण्डी पद्मकस्तिलकस्तथा[262] ॥ ११२ ॥

247. C, D : द्विजादीनां
248. D : काञ्चमालञ्च खदिरं खादिरं च मधूककम्
249. C : राजाटनो; D : राजाटनं
250. D : सहिता
251. D : आङ्गुलविस्तीर्णा
252. D : गुणतापङ्क्तिखातका
253. C, D : उक्त
254. F : क्रमात् instead of तु वा
255. A, C, F : कुटचस्तथा; E : कुटशस्तथा
256. E : बिल्वं
257. E : सनलस्तथा; F : सुरभुस्तथा
258. C : च हयाक्षा-; D : च गयाक्षा-
259. D : कर्णिकश्चैव; E : कण्टकोवृक्ष-
260. C : पारस्कारः
261. F : वरुण-
262. D : मुकुलस्थलकस्तथा

104b-108 These are the other trees which may be used for *muhūrtastambha*, for the gods as well as for the higher castes: *kārtamāla*, *khadira*, *khādira*, *madhūka* and *rājādana*. Herewith the figures for the dimensions (of the *muhūrtastambha*), the height and diameter: the height is twelve, eleven, ten or nine spans and the diameter that same number of digits although, at the summit, this diameter is less by one-tenth; the concealed part (at the bottom) is five, four and a half, four or three spans. The heights and widths prescribed above and calculated according to the number of storeys may also be used. In the case of a *jhaṣālāṅghri* the bottom of the pillar should never be hidden.[44]

109-114 *Aśvattha*, *udumbara*, *plakṣa*, *vaṭavṛkṣa*, *saptaparṇa*, *bilva*, *palāśa*, *kuṭaja*, *pīlu*, *śleṣmātakin*, *lodhra*, *kadamba*, *pārijātaka*, *śirīṣa*, *kovidāra*, *tintriṇīka*, *mahādruma* (?), *śilīndhra*, *sarpamāra*, *śalmalī*, *sarala*, *kiṃsúka*, *arimeda*, *abhaya*, *akṣa*, *amalaka*, *kapittha*, *kaṇṭaka*, *putrajīva*, *ḍuṇḍuka*, *kāraskara*, *karañja*, *varaṇa*, *aśvamāraka*,

44. That kind of pillar rests upon the base (see above vv. 10-11).